7.8 Concept of Vector Magnetic Potential :

MU - Dec. 2006, May 2009, Dec. 2009, Nov. 2010

- The concept of vector magnetic potential can be developed by considering the analogy between the electric field and the magnetic field. Thus, as for the point charge configuration in electric field the electric potential at any point is defined as

$$dV = \frac{dq}{4\pi \in_0 r} \qquad \text{...(A)}$$

- From this equation we have following conclusion :
  - i) The electric potential is directly proportional to its source.
  - ii) The electric potential is inversely proportional to the distance of point from source.
  - iii) The electric potential depends on media.
  - iv) The proportionality constant is $1/4\pi$.
- Analogues to these lines we can have, the concept of magnetic potential based on the following points.
  - i) The magnetic potential is directly proportional to its source. (i.e. differential current element $I\,d\bar{l}$).
  - ii) The magnetic potential is inversely proportional to its distance of point from source. (i.e. r).
  - iii) The magnetic potential depends on media (medium constant is μ)
  - iv) The proportionality constant is $1/4\pi$.

Thus analogoues to mathematical definition of electric potential we can write magnetic potential as

$$d\bar{A} = \frac{\mu I\, d\bar{l}}{4\pi r} \qquad \text{...(B)}$$

7.8.1 Relation between $\bar{B}$ and $\bar{A}$:

- Secondly, we know that $\bar{E} = -\nabla V$, this means that the space derivative of electric potential is equal to one of the electric field quantities which depends on media i.e. electric field intensity.
- But for the magnetic potential which is defined in above equation (B) we can easily observe that the magnetic potential is a vector quantity (due to presence of $d\bar{l}$ which is a vector).
- Thus if we want to equate the space derivative of magnetic vector potential with some magnetic quantity which is a vector then the possible space derivative is curl operation and we can write either.

$$\nabla \times \bar{A} = \bar{H} \quad \text{or} \quad \nabla \times \bar{A} = \bar{B}$$

- But as we have accepted that the magnetic potential depends on media then its space derivative i.e. $\nabla \times \bar{A}$ must depend on media. Out of the two magnetic quantities, i.e. $\bar{B}$ and $\bar{H}$ we have studied that $\bar{H}$ is independent of media and $\bar{B}$ depend on media, thus we can define.

$$\boxed{\nabla \times \bar{A} = \bar{B}} \qquad \ldots(7.8.1)$$

- This is the **relation between $\bar{B}$ and $\bar{A}$.**
- So using Equation (B) we can obtain vector magnetic potential and then using Equation (7.8.1), by taking curl of $\bar{A}$, we obtain $\bar{B}$.

7.8.2 Relation between ψ_m and $\bar{A}$:

This concept of magnetic vector potential is practically useless but many times for deriving the expression of magnetic field intensity for complex fields the concept is widely used.

Vector potential $\bar{A}$ relates to the magnetic flux ψ through a given area s that is bounded by contour C in a simple way.

$$\psi_m = \int_s \bar{B} \cdot d\bar{s}$$

Using Equation (7.8.1) and applying Stoke's theorem

$$\psi_m = \int_s (\nabla \times \bar{A}) \cdot d\bar{s} = \oint \bar{A} \cdot d\bar{l} \text{ (Wb)}$$

i.e.

$$\boxed{\psi_m = \oint \bar{A} \cdot d\bar{l}} \qquad \ldots(7.8.2)$$

7.8.3 Poisson's Equations for Magnetostatic Field :

We have $\bar{B} = \nabla \times \bar{A}$ ...(i)

For the sake of simplicity we choose $\nabla \cdot \bar{A} = 0$ ...(ii)

Taking the curl of Equation (i) we get,

$$\nabla \times \nabla \times \bar{A} = \nabla \times \bar{B} = \nabla \times \mu \bar{H} = \mu \nabla \times \bar{H} = \mu \bar{J} \qquad \ldots(iii)$$

But the vector identity curl of curl of a vector is,

$$\nabla \times \nabla \times \bar{A} = \nabla (\nabla \cdot \bar{A}) - \nabla^2 \bar{A} \qquad \ldots(iv)$$

Using Equations (ii) and (iv), Equation (iii) becomes,

$$\nabla \times \nabla \times \bar{A} = -\nabla^2 \bar{A} = \mu \bar{J}$$

i.e.

$$\boxed{\nabla^2 \bar{A} = -\mu \bar{J}} \qquad \ldots(7.8.2(a))$$

This is **vector Poisson's equation**. In terms of rectangular components of $\bar{A}$ and $\bar{J}$.

$$\nabla^2 \bar{A} = \bar{a}_x \nabla^2 A_x + \bar{a}_y \nabla^2 A_y + \bar{a}_z \nabla^2 A_z = -\mu (J_x \bar{a}_x + J_y \bar{a}_y + J_z \bar{a}_z)$$

Comparing coefficients of $\bar{a}_x$, $\bar{a}_y$ and $\bar{a}_z$, we get

$$\nabla^2 A_x = -\mu J_x \quad \text{...(v)}$$

$$\nabla^2 A_y = -\mu J_y \quad \text{...(vi)}$$

$$\nabla^2 A_z = -\mu J_z \quad \text{...(vii)}$$

Each of these three equations has the same form as the Poisson's equation in electrostatics. These equations are called as Poisson's equations for magnetostatic field. In free space the equation $\nabla^2 V = -\rho_v/\epsilon_0$, has a particular solution,

$$V = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_v}{R} dv$$

Hence, the solution for Equations (v), (vi) and (vii) are

$$A_x = \frac{\mu}{4\pi} \iiint \frac{J_x\, dv}{r} ; \qquad A_y = \frac{\mu}{4\pi} \iiint \frac{J_y\, dv}{r} ; \qquad A_z = \frac{\mu}{4\pi} \iiint \frac{J_z\, dv}{r}$$

Taking the vector sum of these components for $\bar{A}$ gives,

$$\bar{A} = \frac{\mu}{4\pi} \iiint \frac{1}{r} (J_x \bar{a}_x + J_y \bar{a}_y + J_z \bar{a}_z)\, dv$$

i.e.

$$\boxed{\bar{A} = \int_{vol.} \frac{\mu \bar{J}\, dv}{4\pi r}} \quad \text{...(7.8.3)}$$

7.8.4 Solved Examples on Vector Magnetic Potential :

Important Formulae

$$\nabla \times \bar{A} = \bar{B}$$

$$\psi_m = \oint \bar{A} \cdot d\bar{l}$$

Example 7.8.1 : A direct current I flows in a straight wire of length 2L. At a point located at a distance r from the wire in the bisecting plane determine
(i) the vector magnetic potential and (ii) the magnetic flux density.

MU - Nov. 2004

Solution : Consider the current carrying element placed along z-axis with center of the filament at origin as shown in Fig. Ex. 7.8.1 The distance vector $\bar{R}$ of point P from the current element $I\, dz\, \bar{a}_z$ is

$$\bar{R} = r\,\bar{a}_r - z\,\bar{a}_z \text{ and } \bar{a}_R = \frac{r\,\bar{a}_r - z\,\bar{a}_z}{\sqrt{r^2+z^2}}$$

(i) To find vector magnetic potential :

The magnetic vector potential at point P is

$$\bar{A} = \frac{\mu}{4\pi}\int \frac{I\,d\bar{l}}{R}$$

Since $d\bar{l} = dz\,\bar{a}_z$ and assuming free space $\mu = \mu_0$, we have

$$\bar{A} = \frac{\mu_0 I}{4\pi}\bar{a}_z \int_{-L}^{L} \frac{dz}{\sqrt{r^2+z^2}} = \frac{\mu_0 I}{4\pi}\left[\ln z + \sqrt{z^2+r^2}\right]_{-L}^{L} \bar{a}_z$$

$$= \frac{\mu_0 I}{4\pi} \ln\left(\frac{\sqrt{L^2+r^2}+L}{\sqrt{L^2+r^2}-L}\right)\bar{a}_z = A_z\,\bar{a}_z \qquad \text{...Ans.}$$

(ii) To find $\bar{B}$: To find $\bar{B}$ form $\bar{A}$ we have,

$$\bar{B} = \nabla\times\bar{A} = \begin{vmatrix} 1/r\bar{a}_r & \bar{a}_\phi & 1/r\bar{a}_z \\ \partial/\partial r & \partial/\partial\phi & \partial/\partial z \\ 0 & 0 & A_z \end{vmatrix}$$

$$= \frac{1}{r}\bar{a}_r\frac{\partial A_z}{\partial\phi} - \bar{a}_\phi\frac{\partial A_z}{\partial r}$$

$$= -\frac{\partial A_z}{\partial r}\bar{a}_\phi \qquad \text{...Since } A_z \text{ is not a function of } \phi$$

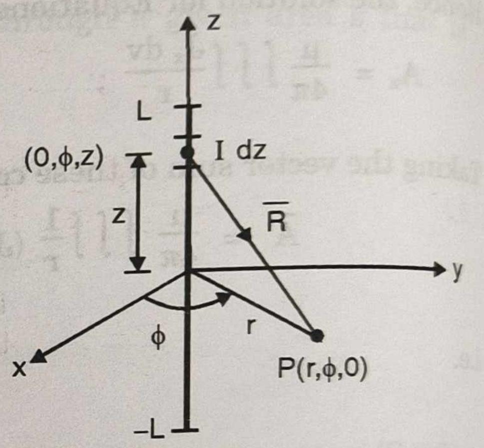


Fig. : Illustrating example 7.8.1

Thus, $$\bar{B} = -\bar{a}_\phi\frac{\partial}{\partial r}\left[\frac{\mu_0 I}{4\pi}\ln\frac{\sqrt{L^2+r^2}+L}{\sqrt{L^2+r^2}-L}\right] = \bar{a}_\phi\frac{\mu_0 IL}{2\pi r\sqrt{L^2+r^2}} \qquad \text{...(A)}$$

When r << L, above equation reduces to $\bar{B} = \frac{\mu_0 I}{2\pi r}\bar{a}_\phi$

which is the expression for $\bar{B}$ at a point located at a distance r from an infinitely long, straight wire carrying current I.

Example 7.8.2 : Using same data in above example find $\bar{B}$ using Biot savart law.

Solution : Applying Biot-savart law :

We have, $$\bar{B} = \mu_0\oint\frac{I\,d\bar{l}\times\bar{a}_R}{4\pi R^2} = \frac{\mu_0 I}{4\pi}\int_{-L}^{L}\frac{dz\,\bar{a}_z}{(r^2+z^2)}\times\frac{r\,\bar{a}_r - z\,\bar{a}_z}{\sqrt{r^2+z^2}} = \frac{\mu_0 I}{4\pi}\int_{-L}^{L}\frac{r\,dz}{(r^2+z^2)^{3/2}}$$

the distance is large compared with dimensions of source (area 'A' and length 'l'), the wire with area 'A' is considered to be very thin wire. It becomes a filamentary conductor, so that we apply the formula,

$$\bar{A} = \frac{\mu}{4\pi}\int \frac{I\,d\bar{l}}{R}$$

Now this problem will be similar to previous problem, only difference is in length of the wire, it is not 2L, it is L. The rest of the solution is left to the students.

Example 7.8.5 : Given the magnetic vector potential.

$\bar{A} = -r^2/4\ \bar{a}_z$ (Wb/m)

Calculate the total magnetic flux crossing the surface

$\phi = \pi/2$, $1 \le r \le 2$m, $0 \le z \le 5$m.

Solution : First we obtain magnetic flux density $\bar{B}$

$$\bar{B} = \nabla \times \bar{A} = \begin{vmatrix} \frac{1}{r}\bar{a}_r & \bar{a}_\phi & \frac{1}{r}\bar{a}_z \\ \partial/\partial r & \partial/\partial\phi & \partial/\partial z \\ 0 & 0 & -r^2/4 \end{vmatrix} = -\bar{a}_\phi \frac{\partial}{\partial r}(-r^2/4) = \frac{r}{2}\,\bar{a}_\phi$$

For the $\phi = \pi/2$ = constant surface,

$$d\bar{s} = dr\,dz\,\bar{a}_\phi$$

Flux through given surface is

$$\psi_m = \int_s \bar{B}\cdot d\bar{s} = \int_s \frac{r}{2}\bar{a}_\phi \cdot dr\,dz\,\bar{a}_\phi = \int_0^5\int_1^2 \frac{r}{2}\,dr\,dz = \frac{1}{2}\left(\frac{r^2}{2}\right)_1^2, (z)_0^5 = 3.75\ \text{(Wb)}$$

7.9 Concept of Scalar Magnetic Potential :

MU - May 2009, Dec. 2009, Nov. 2010

We have, $\quad \nabla \times \bar{A} = \bar{B}$

taking curl of both sides and knowing that $\bar{B} = \mu\bar{H}$ and $\nabla \times \bar{H} = \bar{J}$,

$$\nabla \times \nabla \times \bar{A} = \nabla \times \bar{B}$$

$$= \mu(\nabla \times \bar{H})$$

i.e. $$\nabla \times \bar{B} = \mu\bar{J} \quad \text{...(7.9.1)}$$

In current free region $\bar{J} = 0$ gives

$$\nabla \times \bar{B} = 0 \quad \text{...(7.9.2)}$$

The vector identity is

Curl of gradient = 0

In Equation (7.9.2), since $\nabla \times \bar{B} = 0$, now $\bar{B}$ can be expressed as gradient of a scalar field. Let

$$\bar{B} = -\mu_0 \nabla V_m \quad ...(7.9.3)$$

where, V_m is called the scalar magnetic potential (expressed in amperes). The negative sign is conventional and the permeability of free space μ_0 is simply proportionality constant. In electrostatics we have

$$V_2 - V_1 = \int_{P_1}^{P_2} \bar{E} \cdot d\bar{l}$$

Similarly the scalar magnetic potential difference can be expressed as

$$V_{m2} - V_{m1} = -\int_{P_1}^{P_2} \frac{1}{\mu_0} \bar{B} \cdot d\bar{l} \quad ...(7.9.4)$$

If there were magnetic charges with a volume density ρ_m (A/m²) in a volume v', then V_m can be obtained by using,

$$V_m = \frac{1}{4\pi} \int_{V} \frac{\rho_m}{R} dv' \text{ (A)} \quad ...(7.9.5)$$

- The magnetic flux density $\bar{B}$ could then be determined from equation (7.9.3). However, isolated magnetic charges have never been observed experimentally, they must be considered fictitious.
- The magnetic field of a small bar magnet is the same as that of magnetic dipole. This can be verified experimentally be observing contours of iron filings around a magnet. The ends (the north and south poles) of a permanent magnet are the location of positive and negative magnetic charges respectively.
- For a bar magnetic the fictitious magnetic charges $+q_m$ and $-q_m$ are assumed to be separated by a distance d to form an equivalent magnetic dipole of moment.

$$\bar{m} = q_m \bar{d} = \bar{a}_n \text{ I s} \quad ...(7.9.6)$$

The scalar magnetic potential is then

$$V_m = \frac{\bar{m} \cdot \bar{a}_R}{4 \pi R^2} \text{ (A)} \quad ...(7.9.7)$$

- This relation is exactly analogous to that of the electric scalar potential for an electric dipole.
- It is to be noted that the curl free nature of $\bar{B}$ ($\nabla \times \bar{B} = 0$) from which the scalar magnetic potential is developed, holds only at points with no currents.
- In a region where current exist, the magnetic field is not conservative, and the scalar magnetic potential is not a single valued function, hence the magnetic potential difference evaluated by Equation (7.9.4) depends on the path of integration.

- For these reasons we will use the circulating current and vector potential approach instead of the fictitious magnetic charge and scalar potential approach.

7.10 Magnetic Boundary Conditions :

MU - May 2010

When the field vector is present in a single continuous media then at every point in a media, the magnitude and direction of field will be same. When light travels from one media to second media, the direction of light changes. Similar to this when the field goes from one media to second media, the direction as well as magnitude of field changes.

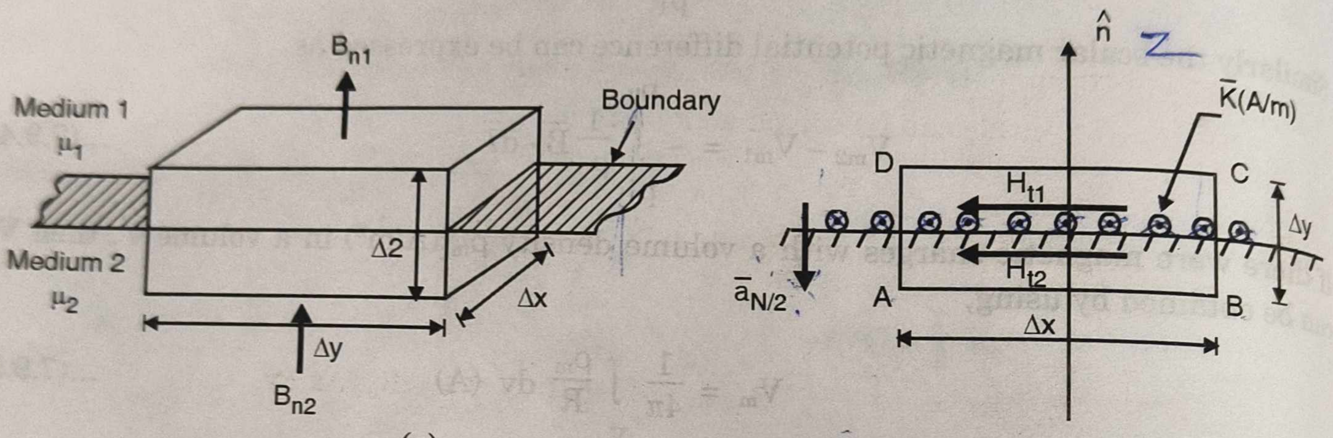


Fig. 7.10.1 : To determine boundary conditions

Suppose the field in one media is given and we require to calculate field in second media. This type of problems are solved by knowing the relation between the tangential and normal components of the fields in two media. As in electrostatics, the relation between tangential and normal components is derived separately.

Fig. 7.10.1. shows a boundary between two isotropic homogeneous linear medias with permeabilities μ_1 and μ_2. The gaussian surface and the closed path is constructed at the boundary between medium 1 and medium 2 as shown.

7.10.1 Boundary Condition for Normal Components of $\bar{B}$:

Consider a small rectangular volume positioned across the interface such that it is half in each media. Instead of rectangular volume any closed surface can be considered. The normal components of $\bar{B}_1$ and $\bar{B}_2$ are B_{n1} and B_{n2} present at the top and bottom of the volume. Since magnetic flux lines are continuous we have,

$$\oint \bar{B} \cdot \overline{dS} = 0$$

The closed surface consists of six surfaces. Thus L.H.S of above equation is obtained by

$$\text{LHS} = \oint \bar{B} \cdot \overline{dS} = \int_{\text{Top}} \bar{B} \cdot \overline{dS} + \int_{\text{Bottom}} \bar{B} \cdot \overline{dS} + \int_{\text{Left}} \bar{B} \cdot \overline{dS} + \int_{\text{Right}} \bar{B} \cdot \overline{dS} + \int_{\text{Front}} \bar{B} \cdot \overline{dS} \int_{\text{Back}} \bar{B} \cdot \overline{dS}$$

To find the relation between normal components at the interface, B_{n1} and B_{n2} must be very close to the interface which requires,

$$\Delta z \rightarrow 0$$

The effect of it is the contribution of all sides except top and bottom is zero and now LHS consists of only two integrals.

$$\therefore \quad \text{LHS} = \int_{\text{Top}} \bar{B} \cdot \overline{dS} + \int_{\text{Bottom}} \bar{B} \cdot \overline{dS}$$

At the top : $\bar{B} = B_{n1}\, \bar{a}_z$; $\overline{dS} = dx\, dy\, \bar{a}_z$

At the bottom : $\bar{B} = B_{n2}\, \bar{a}_z$; $\overline{dS} = -dx\, dy\, \bar{a}_z$

Now the integrals can be solved as follows :

$$\int_{\text{Top}} \bar{B} \cdot \overline{dS} = \int_0^{\Delta y} \int_0^{\Delta x} B_{n1}\, \bar{a}_z \cdot dx\, dy\, \bar{a}_z = B_{n1}\, \Delta x\, \Delta y$$

$$\int_{\text{Bottom}} \bar{B} \cdot \overline{dS} = \int_0^{\Delta y} \int_0^{\Delta x} B_{n2}\, \bar{a}_z \cdot dx\, dy\, (-\bar{a}_z) = -B_{n2}\, \Delta x\, \Delta y$$

$$\therefore \quad \text{LHS} = (B_{n1} - B_{n2})\, \Delta x\, \Delta y = \text{RHS} = 0$$

$$\therefore \quad B_{n1} = B_{n2} \qquad \text{...(7.10.1)}$$

i.e. normal component of $\bar{B}$ is continuous across an interface.

7.10.2 Boundary Conditions for Tangential Components of $\bar{H}$:

Let H_{t1} and H_{t2} are the tangential components of H_1 and H_2 in medium 1 and medium 2 respectively. To obtain these boundary relations consider a rectangular path half in each media with width Δx and height Δy as shown in Fig. 7.10.1(b). We have Amperes work law –

$$\oint \bar{H} \cdot d\bar{l} = I_{\text{enclosed}} \qquad \text{.....(A)}$$

In the figure, the circle with dot indicates that the current is coming out of the page. This current is the surface current (K) flowing along the interface. Since width of the rectangle is Δx, the current enclosed within the path is $K\Delta x$

$$\therefore \text{RHS} = I_{\text{enclosed}} = K\Delta\, x$$

The rectangular path consists of four sides, thus LHS of eq. (A) can be split as

$$\text{LHS} = \oint \bar{H} \cdot d\bar{l} = \int_A^B \bar{H} \cdot d\bar{l} \int_B^C + \int_C^D + \int_D^A$$

If we want H_{t1} and H_{t2} to be present very close to the interface, let

$$\Delta y \to 0$$

The effect of it is integrals B to C and D to A gives zero contribution to LHS. Thus LHS reduces to,

$$\text{LHS} = \int_A^B \bar{H}\cdot d\bar{l} + \int_C^D \bar{H}\cdot d\bar{l}$$

For side A to B, $\bar{H}$ and $d\bar{l}$ are in same direction, thus

$$\int_A^B \bar{H}\cdot d\bar{l} = H_{t1}\int_A^B dl = H_{t1}\,\Delta x$$

For side C to D, $\bar{H}$ and $d\bar{l}$ are in opposite direction, thus

$$\int_C^D \bar{H}\cdot d\bar{l} = -H_{t2}\int_C^D dl = -H_{t2}\,\Delta x$$

LHS of eq (A) now becomes

$$\text{LHS} = (H_{t1} - H_{t2})\,\Delta x = \text{RHS} = K\Delta x$$

$$\text{i.e.} \quad \boxed{H_{t1} - H_{t2} = K} \qquad \text{...(7.10.2)}$$

In vector form,

$$\bar{H}_{t1} - \bar{H}_{t2} = \bar{a}_{N12} \times \bar{K} \qquad \text{...(7.10.3)}$$

Where, $\bar{a}_{N12}$ is the unit vector normal to interface from medium 1 to medium 2.

If no current along interface, then

$$H_{t1} - H_{t2} = 0$$

$$\text{or} \quad \boxed{H_{t1} = H_{t2}} \qquad \text{...(7.10.4)}$$

Example 7.10.1 : Two homogeneous, linear and isotropic media have an interface at $x = 0$. $x < 0$ describes medium 1 and $x > 0$ describes medium 2. $\mu_{r1} = 2$ and $\mu_{r2} = 5$. The magnetic field in medium 1 is

$150\,\bar{a}_x - 400\,\bar{a}_y + 250\,\bar{a}_z$ A/m.

Determine

i) Magnetic field in medium 2.

ii) Magnetic flux density in medium 1.

iii) Magnetic flux density in medium 2.

MU - June 2005

...(ii)

Solution :

(i) **To obtain $\bar{H}_{t2}$:**

Since x-axis is normal to the interface, the normal and tangential components of :

$$\bar{H}_{t1} = -400\ \bar{a}_y + 250\ \bar{a}_z\ \text{(A/m)}$$

$$\bar{H}_{n1} = 150\ \bar{a}_x\ \text{(A/m)}$$

From the boundary condition we have,

$$\bar{H}_{t2} = \bar{H}_{t1} = -400\ \bar{a}_y + 250\ \bar{a}_z$$

To obtain normal component, we have

$$\bar{B}_{n1} = \mu_1 \bar{H}_{n1}$$

$$= 2\ \mu_o (150\ \bar{a}_x)$$

$$= 300\ \mu_o\ \bar{a}_x\ \text{(Wb/m}^2\text{)}$$

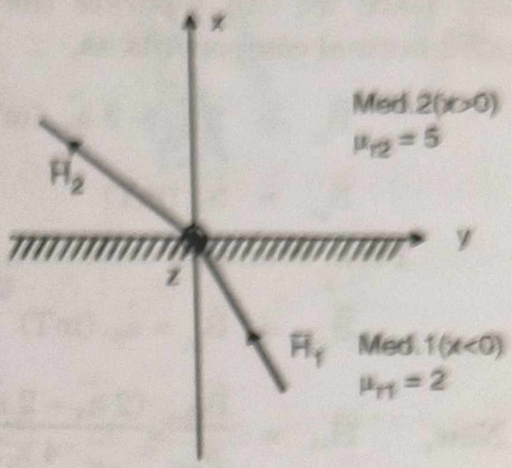


Fig. : Illustrating Ex.7.10.1

From the boundary condition we have,

$$\bar{B}_{n2} = \bar{B}_{n1} = 300\ \mu_o\ \bar{a}_x\ \text{(Wb/m}^2\text{)}$$

we have,

$$\bar{H}_{n2} = \frac{\bar{B}_{n2}}{\mu_2} = \frac{300\ \mu_o\ \bar{a}_x}{5\ \mu_o} = 60\ \bar{a}_x\ \text{(A/m)}.$$

Thus, the magnetic field in medium 2 is,

$$\bar{H}_2 = \bar{H}_{n2} + \bar{H}_{t2} = 60\ \bar{a}_x - 400\ \bar{a}_y + 250\ \bar{a}_z\ \text{(A/m)}.$$

(ii) **To find $\bar{B}_1$:** we have,

$$\bar{B}_1 = \mu_1 \bar{H}_1 = 2\ \mu_o (150\ \bar{a}_x - 400\ \bar{a}_y + 250\ \bar{a}_z)$$

$$= 188.5\ \bar{a}_x - 502.7\ \bar{a}_y + 314.16\ \bar{a}_z\ (\mu\ \text{Wb/m}^2)$$

(iii) **To find $\bar{B}_2$:** we have,

$$\bar{B}_2 = \mu_2 \bar{H}_2 = 5\ \mu_0 (60\ \bar{a}_x - 400\ \bar{a}_y + 250\ \bar{a}_z)$$

$$= 377\ \bar{a}_x - 2513\ \bar{a}_y + 1570\ \bar{a}_z\ (\mu\ \text{Wb/m}^2)$$

Example 7.10.2 : Let $\mu 1 = 4\ \mu$H/m in the region 1 where $z > 0$ while $\mu_2 = 7\ \mu$H/m wherever $z < 0$. In region 1 the magnetic flux density

$$\bar{B}_1 = 2\ \bar{a}_x - 3\ \bar{a}_y + \bar{a}_z\ \text{(mT)},\ \bar{K} = 80\ \bar{a}_x\ \text{A/m}$$

on the surface $z = 0$. Find the value of $\bar{B}_2$ in region 2.

Solution :

Given : $\bar{B}_1 = 2\,\bar{a}_x - 3\,\bar{a}_y + \bar{a}_z$ (mT)

Since z-axis is normal to the interface we can separate the tangential and normal components as,

$$\bar{B}_{t1} = 2\,\bar{a}_x - 3\,\bar{a}_y \text{ (mT)}$$

$$\bar{B}_{n1} = \bar{a}_z \text{ (mT)}$$

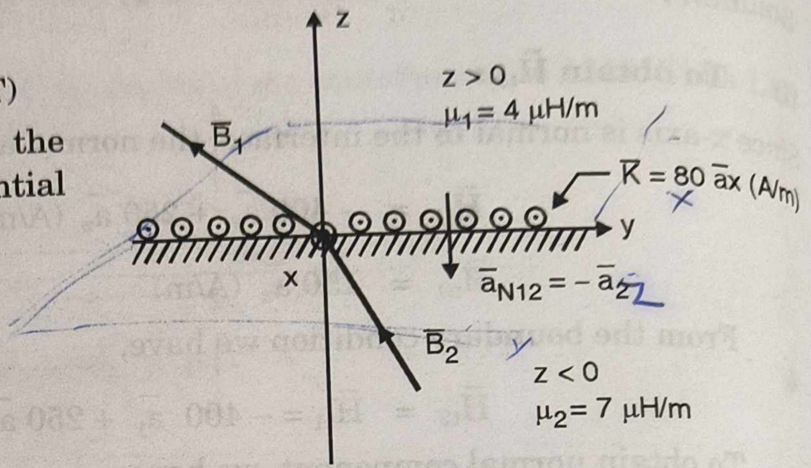


Fig. : Illustrating Ex. 7.10.2

From boundary condition,

$$\bar{B}_{n2} = \bar{B}_{n1} = \bar{a}_z \text{ (mT)}$$

Now, $$\bar{H}_{t1} = \frac{\bar{B}_{t1}}{\mu_1} = \frac{(2\,\bar{a}_x - 2\,\bar{a}_y) \times 10^{-3}}{4 \times 10^{-6}} = 500\,\bar{a}_x - 750\,\bar{a}_y \text{ (A/m)}$$

From boundary condition,

$$\bar{H}_{t1} - \bar{H}_{t2} = \bar{a}_{N/2} \times \bar{K}$$

$$\therefore\ \bar{H}_{t2} = \bar{H}_{t1} - (\bar{a}_{N/2} \times \bar{K}) = (500\,\bar{a}_x - 750\,\bar{a}_y) - (-\bar{a}_z \times 80\,\bar{a}_x)$$

$$= 500\,\bar{a}_x - 670\,\bar{a}_y \text{ (A/m).}$$

The tangential component of $\bar{B}_{t2}$ is

$$\bar{B}_{t2} = \mu_2\,\bar{H}_{t2} = (7 \times 10^{-6}) \times (500\,\bar{a}_x - 670\,\bar{a}_y) = 3.5\,\bar{a}_x - 4.69\,\bar{a}_y \text{ (mT).}$$

Thus the magnetic flux density in medium 2 is

$$\bar{B}_2 = \bar{B}_{t2} + \bar{B}_{n2} = 3.54\,\bar{a}_x - 4.69\,\bar{a}_y + \bar{a}_z \text{ (mT).}$$

Example 7.10.3 : Show that $\dfrac{\tan\alpha_1}{\tan\alpha_2} = \dfrac{\mu_{r1}}{\mu_{r2}}$.

Solution : Consider the interface between two magnetic media with finite conductivities. Let α_1 and α_2 are the angles made with the normal by the magnetic fields in region 1 and region 2, respectively, as shown in figure. As the normal components of $\bar{B}$ are continuous we obtain

$$B_1 \cos\alpha_1 = B_2 \cos\alpha_2$$

Since each medium has a finite conductivity (assumption), $\bar{K} = 0$. Therefore tangential components of the $\bar{H}$ field, are continuous. That is

$$H_{t1} = H_{t2}$$

$$\text{or}\quad \frac{B_{t1}}{\mu_1} = \frac{B_{t2}}{\mu_2} \quad\text{or}\quad \frac{B_1 \sin\alpha_1}{\mu_1} = \frac{B_2 \sin\alpha_2}{\mu_2}$$

7.11 Inductance-Self and Mutual :

Self Inductance :

Consider a coil with N closely wound turns connected to a time varying source and carrying a changing current I as shown in the Fig. 7.11.1. The changing current will produce a changing magnetic flux to link with the circuit and hence, by Faraday's law of induction, will induce an emf in the circuit to oppose the change in flux i.e. the change of current. The emf so induced is called the self induced emf (self-because the changing flux Ψ_m is produced by the current in the loop itself.), while the phenomena is known as self induction and the property of the circuit to produce self induction is called self inductance (Faraday's law is explained in detail in next chapter).

The product $N\Psi_m$ is commonly called the number of flux linkages and is denoted by λ :

$$\lambda = N\Psi_m$$

The self inductance of a circuit is now mathematically defined as

$$L = \frac{d\lambda}{dI}$$

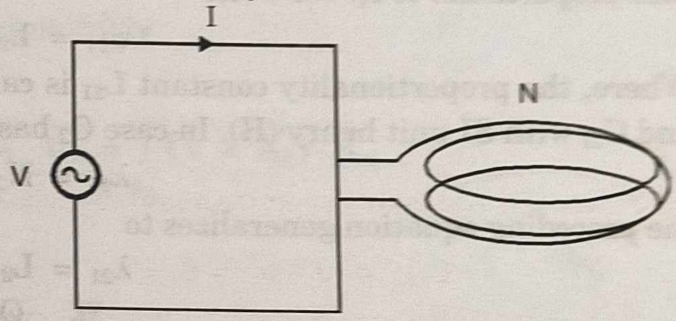


Fig. 7.11.1 : Defining self inductance

Where, L is the self inductance of the circuit in Wb/A or H.

From the Biot-Savart law, we see that flux density B is directly proportional to current I, hence flux ψ_m is also proportional to I. We write total flux linkage.

$$N\psi_m = L\,I \qquad \text{or } \lambda = L\,I \qquad \text{or } L = \frac{\lambda}{I} \text{ (H)}$$

for a linear medium. In general $L = \frac{d\lambda}{dI}$ (H).

Since the induced emf from the Faraday's law :

$$e = -\frac{d\lambda}{dt} = -\frac{d}{dt}(LI)$$

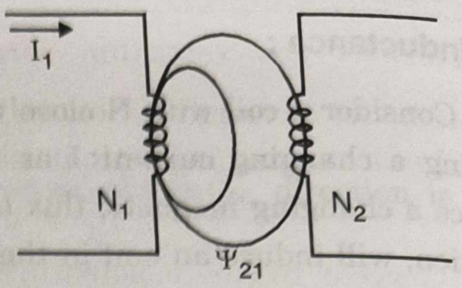
$$\boxed{e = -L\frac{dI}{dt}}$$

Thus, the self inductance of a circuit is 1 henry if an emf of 1 volt is induced in it when the current in it changes at the rate of 1 amp per second.

Mutual Inductance :

If instead of considering only one circuit, we consider two neighbouring closed loops, C_1 and C_2 bounding surfaces S_1 and S_2 respectively, as shown in Fig. 7.11.2.

A current I_1 in C_1 sets up a magnetic field $\bar{B}_1$. Some of the magnetic flux due to $\bar{B}_1$ will link with C_2, i.e., will pass through the surface S_2 bounded by C_2. Here, the concept of mutual inductance comes into picture. The flux linked with C_2 is termed as **mutual flux,** denoted by ψ_{21}. We have



$$\psi_{21} = \int_{S_2} d\bar{s} \text{ (Wb)}$$

Fig. 7.11.2 : Set-up for finding mutual inductance

Again from Bio-Savart law, we see that $\bar{B}_1$ is directly proportional to I_1, hence ψ_{21} is also proportional to I_1. We write

$$\psi_{21} = L_{21} I_1$$

Where, the proportionality constant L_{21} is called the mutual inductance between loops C_1 and C_2, with SI unit henry (H). In case C_2 has N_2 turns, the flux linkage λ_{21} due to ψ_{21} is

$$\lambda_{21} = N_2 \psi_{21} \text{ (wb)},$$

the preceding equation generalizes to

$$\lambda_{21} = L_{21} I_1$$

Or

$$\boxed{L_{21} = \frac{\lambda_{21}}{I_1} \text{ (H)}} \quad \text{...(7.11.1)}$$

In the above discussion it is assumed that the permeability of the medium does not change with I_1, i.e. above equations are valid only to linear media. A more general definition for L_{21} is

$$L_{21} = \frac{d\lambda_{21}}{dI_1} \text{ (H)}$$

Now if e_2 is the emf induced in circuit 2 due to the linkages λ_{21} then

$$e_2 = -\frac{d\lambda_{21}}{dt}$$

Substituting for $d\lambda_{21}$ from the preceding equation

$$e_2 = -L_{21} \frac{dI_1}{dt}$$

On the similar lines we may e_1 when current I_2 flowing in C_2,produces flux linkage λ_{12} :

$$e_1 = -L_{12} = \frac{d I_2}{dt}$$

For a linear system $\quad L_{12} = L_{21}$

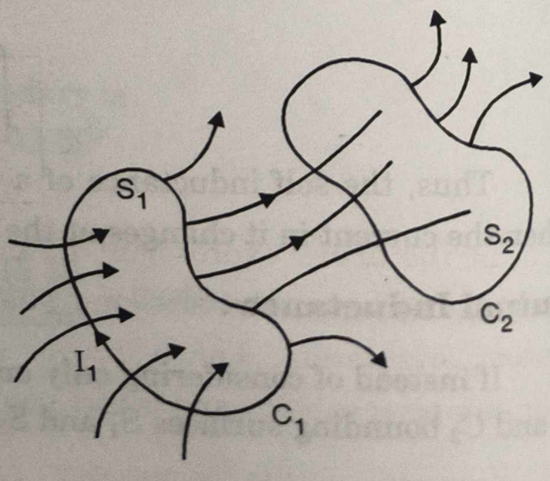


Fig. 7.11.3 : Two magnetically coupled loops